

Motion and Whiplash Effect of a Floating Crane Model Under Wave Load: Experiment

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ABSTRACT

The top parts of tall terrestrial buildings are sometimes damaged or even destroyed during strong earthquakes because of the whiplash effect. Similarly, the marine floating high-rise multibody structure with a floating base may experience the whiplash effect even more easily, which means that marked motion of the upper part of the structure will occur accompanying minor motion of the floating base under the action of an external load induced by a wave or seismic load. The whiplash effect would threaten the operation and the survival of these offshore high-rise platform structures.

In this paper, a physical experiment involving a floating crane without a mooring system under wave loads was set up. The data and results were analyzed and discussed mainly in terms of the motion and the whiplash effect of the floating crane. The whiplash coefficient is defined to estimate the magnitude of the whiplash effect. Some factors, including wave amplitude, wave length, boom angle and boom length, are investigated to determine how they affect the motion and the whiplash effect of the floating crane. Some suggestions are given to control the whiplash

effect, and those are useful for the operation of offshore floating platform structures, such as the oil-drill platform, the floating wind farm, the floating crane and some other high-rise marine floating structures.

ADDITIONAL INDEX WORDS: *Floating crane, Multibody system dynamics, Whiplash effect.*

INTRODUCTION

There are various types of marine floating platforms that are commonly used in coastal and deep-sea regions. The dynamic problems associated with these structures differ from the problems arising for terrestrial buildings or offshore structures with fixed piles. In some cases, the dynamic behavior of the marine floating structures is considered critical with respect to the amplitudes of the motion of the vessel or the load. Even small disturbances in the state of the system, such as a high-rise floating crane under a wave load, can cause an evident dynamic response in the upper parts, such as the motion of heavy cargo. In addition, the amplitude of the motion of the hull has to remain small to achieve the required positioning accuracy, to avoid damages at the mooring system, to prevent destruction resulting from the whiplash effect, etc.

The investigation of floating cranes operating offshore has been of interest for many years. Schellin *et al.* (1993) set up the equation of motion for considering the 6-degrees-of-freedom floating crane and the translational motions of the 3-degrees-of-freedom cargo along the X , Y , and Z axes. The interaction between the floating crane and the cargo is considered, and the wire rope is extensible. The hydrostatic and hydrodynamic forces, the viscous damping force, the mooring force, and the wire rope force are considered as the external forces. Ellermann *et al.* (2002, 2005) assume that the floating crane and the cargo move in a 2-dimensional plane, and they consider the 3-degrees-of-freedom motion of the floating crane. The nonlinear equation of motion is set up for considering the large amplitude motions of the floating crane and the cargo based on multibody system dynamics. The solution is obtained analytically, and the nonlinearity

is studied. Cha *et al.* (2010) investigated the dynamic response simulation of heavy cargo suspended by a floating crane. The dynamic equations of motion are set up for considering the 6-degrees-of-freedom floating crane and the 6-degrees-of-freedom cargo based on multibody system dynamics. The nonlinear terms in the equations of motion are considered. In addition, linear hydrostatic force, the nonlinear hydrodynamic force, the wire rope force, and the mooring force are considered as the external forces. Han *et al.* (2011) establish a procedure for the systematic design and performance evaluation of an offshore platform crane using a computational approach. With the development of deep-sea exploitation in the future, the floating crane will be one of the essential facilities used for engineering applications.

The original meaning of whiplash effect stems from the idea that the velocity of a whip tail would approach or exceed the sound velocity and give rise to a sharp sound because of the super-wave sparked by the whiplash if one moves the whip with a certain frequency. In terrestrial building construction, the whiplash effect will occur in the upper parts of high-rise buildings subjected to seismic loads. The upper parts could be seriously damaged or destroyed even if the main building still stands.

Zhong & Lin (1985) explained the mechanism of the whiplash effect in a simple manner by using the oscillation of coupling vibrators, and they suggested that adding the stiffness of the upper parts of tall buildings does not always eliminate the whiplash effect but does change the distribution of the mass and stiffness, as determined through analysis and numerical calculation. Li & Zhong (2010) investigated the dynamic characteristics of a special super high-rise steel TV tower and analyzed the response of the building under frequent earthquakes using both the response spectral analysis and time-history analysis, respectively. The seismic response of the main structure is larger using spectral analysis, whereas by resorting to a time-history analysis, the whiplash effect can be clearly determined.

Papers considering the dynamics of the floating multibody system seldom discuss the whiplash effect. There are more complex external loads, including wave, wind, current, seismic, and collision loads, acting on the floating multibody system than on terrestrial high-rise buildings. Zhong & Lin (1985) pointed out that the whiplash effect would also take place in offshore platform structures, not only in terrestrial high-rise buildings. In this paper, the dynamic motion of a floating crane subjected to wave loads with different structure parameters is tested and calculated. Then, we focus on the whiplash effect of this floating multibody system. We redefine the whiplash effect for the rigid floating multibody system as follows: the cargo on the floating crane will multiply its swing motion compared with the roll or pitch of the floating base subjected to a wave load. This means that the energy would transfer from the floating base to the heavy cargo at some point and cause the cargo to increase its angle displacement and velocity.

EXPERIMENTAL SETUP

Experimental setup was used to determine the influence of different parameters on the dynamics of the floating crane with heavy cargo at the Jiangsu Science and Technology University. The main subject of interest was the motion and the existence of the whiplash effect. The mooring system was neglected because it is very difficult to investigate its influence on the dynamics precisely.

The floating crane system was combined with four parts, namely, the floating base, the partial weight, the boom and the heavy cargo. The main parameters are listed in Table 1. Each part of the system, which are all cuboid, is uniform in mass distribution. Therefore, it is easy to calculate the moments of inertia, gravity centers and other physical properties. Figure 1 shows a sketch of the setup. Experimental measurements were made in a wave tank that is 100m long and 6m width, with a depth of 2m. Waves are generated by a wavemaker, and the wave frequency range is 0.2~2Hz. The maximum wave amplitude is 0.35m. The amplitude of waves is measured by wave gauges. The wave amplitude refers to the initial calm water level. The measurement region

with the floating crane and the measurement equipment are situated in the middle of the wave tank. On the other end of the wave tank, a porous beach is installed to dissipate the wave and minimize the reflection waves.

Six degrees of freedom of the system motions, including the position and the orientation, are measured by a noncontact measurement instrument. Six light-emitting diodes were placed at the floating crane, three at the floating base and the others at the heavy cargo. The light trajectories were recorded by the optical sensors. The data acquisition and analysis systems are shown in Figure 2.

The investigated floating crane is a simplified model of a crane barge without a mooring system. It is noted that the small scale makes quantitative analysis difficult because the disturbances from the vibrations in the experimental setup or the reflections of waves have a large effect on the dynamics of the floating crane system.

The body-fixed coordinate system is shown in Figure 3. There are 6DOF of motion, namely, surge, sway, heave, roll, pitch and yaw, for the floating base. Another three independent degrees of freedom for the heavy cargo are all angle displacements.

Various parameters affecting the motion of the floating crane are investigated, including the wave length, wave amplitude and the orientation of the boom, i.e., the boom angle ψ , as shown in Figure 4. Five boom angle cases including $\psi = 0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ$ are chosen in the experimental studies.

The boom can rotate about the vertical axis (z-axis), as shown in Figure 5. The boom angles ψ are $0^\circ, 45^\circ$ and 180° in the snapshots shown in (a)~(c), respectively. The measurement is taken before the start of the wave flap, but the data analysis is carried out when the motion of the floating base is steady.

RESULTS

Various Boom Angles

Two different waves are considered in the present study. In case 1, the wave length is 3m, and the wave amplitude is 0.04m; in case 2, the wave length is 4m, and wave amplitude is 0.06m. Both of these waves are linear.

Comparisons of the roll motion of the floating crane are shown in Figure 6. Unfortunately, some of the data for the boom angle of 45° are missing in Figure 6(b). The existing data can show the trend for the rolling motion of the floating base as a function of the boom angle. The case of a boom angle of $\psi = 0^\circ$, the partial load is located on the port side and the heavy load is located on the starboard side of the floating base. In calm water, the structure has an upright position without heel and trim. in the vicinity of rolling angle $\psi = 0^\circ$ if there is no error in the initial conditions and no distribution of boundary conditions. There is no coupling relationship between roll and pitch. The amplitude of the roll motion reaches the maximum value.

When the boom angle ψ is equal to 90° , the floating crane is not in the righting state because of the inclining of the gravity center. The plane containing the boom and the heavy load is vertical to the wave direction. This position has the smallest effect of wave on the rolling motion.

The motion of the floating crane subjected to the wave is a forced movement, so the motion period of the linear system is equal to the wave period. angle amplitude of the floating crane in the waves is more violent if the wave steepness is larger and the wave period is shorter.

Different Wave Cases

Figure 7 is the time series of the roll of the floating crane at different wavelengths and wave amplitudes with various boom angles from 0° to 180° at an interval of 45° . Unfortunately, some of the data for case 2 are missing in Figure 7(d) in the time range [9, 12] minute, but this does not significantly influence the analysis of the result. It is found that the amplitude of the rolling motion in case 1 is larger than in case 2 because the natural frequency of the floating crane is closer to the incident wave for case 1.

When the boom angle is equal to 0° and 180° , the plane containing the boom and the heavy cargo is parallel to the xoz plane. Therefore, the motion of the floating base is significantly affected by the motion of the heavy load. The gravity center of the floating system is inclining to the left side, so the obliquity is the largest and this case is the worst one for the floating crane in operation.

From Figure 7(d), we can determine that the roll amplitude of the floating base under wave case 1 is less than that under case 2. The reason is that the heavy load has more effect than the wave load on the roll motion and the pitching motion is affected by the wave load.

Motion Range of the Heavy Cargo

Figure 8 shows the maximum and minimum values of x and y displacement, *i.e.*, the projected area in the floating base for the heavy cargo's motions. The range in case 1 at boom angles of 0° , 90° and 180° are larger than the range in case 2 at homological boom angles.

When the boom angles are equal to 0° and 180° , the x values are not equal to zero because the initial conditions of pitch angle are not equal to zero or some disturbances exist to generate the pitch motions.

In engineering practice, it is difficult to predict the range of motion because of the coupling motion of roll and pitch. Chaos and bifurcation may exist at some conditions (Ellermann, 2002), and it is important to understand the nonlinear dynamic behavior of the floating crane.

Cargo Effect on the Coupling Motion of Roll and Pitch

Figures 9 and 10 are the comparison of the roll and pitch motion of case 1 and case 2 at various boom angles. The main focus of this type of analysis is how the heavy cargo affects the roll and pitch motion in periodic waves.

From Figures 9(a) and 10(a) and Figures 9(e) and 10(e) at boom angles of 0° and 180° , we can see that the amplitude of the roll motion is much more than that of the pitch motion because the heavy cargo mainly affects the roll motion, whereas the heavy cargo mainly affects the pitch motion in Figures 9(c) and 10(c). When the boom angles are 45° and 135° in case 1 and case 2 in

Figures 9(b), 10(b) and Figures 9(d), 10(d), respectively, the amplitude of the roll motion and pitch motion are equivalent because they are coupled tightly.

DISCUSSION

We define the angle displacement as θ_1 for the floating base of the roll or pitch motion and θ_2 for the heavy cargo. The angle amplitude of each degree is defined as $\theta_a = \theta_{max} - \theta_{min}$. Define the whiplash coefficient W_c for rolling and pitch motion as

$$W_c = \frac{\theta_{a,1}}{\theta_{a,2}}. \quad (1)$$

The yaw motion is not evident and can be neglected during the whiplash effect analysis.

Figure 11 shows the roll amplitude at various boom angles. The roll amplitude reaches the largest value as the boom angle equals 0° , whereas it is the smallest when the boom angle is 90° in both wave cases. The variation for pitch amplitude is opposite, as shown in Figure 12.

The roll and pitch whiplash coefficients are shown in Figures 13 and 14, respectively. From the two figures, we can see that the whiplash coefficients are always larger than 2, even reaching 40. This means that the angle amplitude of the heavy cargo is larger than that of the floating base. In some cases, the small angle displacement of the floating base will induce large angle displacement of the heavy cargo, *i.e.*, the whiplash effect occurs.

CONCLUSIONS

A physical experiment is set up to investigate the motion of a marine floating crane with heavy cargo. The motions of the floating base system are analyzed with various boom angles and different wave parameters. Then, the whiplash coefficients of roll and pitch are investigated.

The experimental results indicate that the roll amplitudes reach their maximum value when the boom angles are equal to 0° and 180° . In contrast, the pitch amplitude reaches its maximum value when the boom angle is equal to 90° under different wave parameters.

A large whiplash coefficient indicates that the whiplash effect occurs. The roll and pitch whiplash coefficients are all larger than 2, even reaching values greater than 40 in the physical

experimental results. This means that a small roll or pitch motion will induce a large swing motion of the heavy cargo.

In engineering practice involving marine multibody systems with a floating base, a small dynamic angle displacement would induce a large displacement of the heavy cargo in the operation of a floating crane. The whiplash coefficient suggested in this paper could measure the coupling effect of the dynamics of the multibody system with the floating base. The whiplash effect would occur when the whiplash coefficient is large. Moreover, the whiplash effect would cause some difficulties in the operation of the marine floating base system, even damaging or destroying the top part of the floating system.

ACKNOWLEDGEMENTS

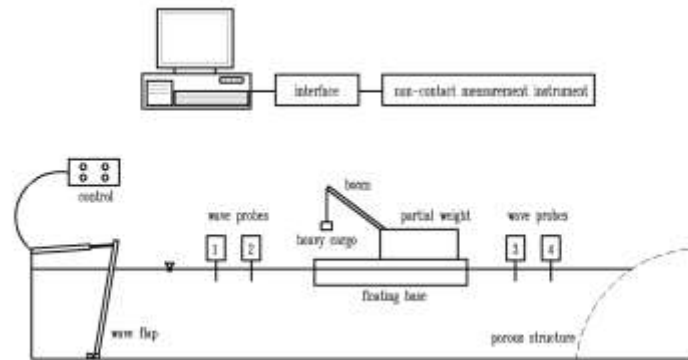
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LITERATURE CITED

- Cha, J.H.; Roh, M.I.; Lee, K.Y., 2010. Dynamic response simulation of a heavy cargo suspended by a floating crane based on multibody system dynamics. *Ocean Engineering*, 37, 1273-1291.
- Ellermann, K., 2005. Dynamics of a moored barge under periodic and randomly disturbed excitation. *Ocean Engineering*, 32, 1420-1430.
- Ellermann, K.; Kreuzer, E.; & Markiewicz M., 2002. Nonlinear dynamics of floating cranes. *Nonlinear Dynamics*, 27, 107-182.
- Han, D.S.; Yoo, S.W.; Yoon, H.S.; *et al.*, 2011. Coupling analysis of finite element and finite volume method for the design and construction of FPSO crane. *Automation in Construction*, 20, 368-379
- Li, C.; Zhong Q., 2010. Numerical simulation of seismic responses of complex shape super high-rise steel TV tower. *Computer Modeling and Simulation*, 1, 14-18.
- Schellin, T.E.; Jiang, T.; Ostergaard, C., 1993. Response analysis and operating limits of crane ships. *Journal of Ship Research*, 37(3), 225-238.
- Zhong, W.X.; Lin, J. H., 1985. "Whiplash effect" of high buildings caused by earthquakes. *Journal of Vibration & Shock*, 2, 1-6. (In Chinese)

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FIGURES



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Figure 1. Experimental setup.

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Figure 2. Measurement instrument and data collection system.

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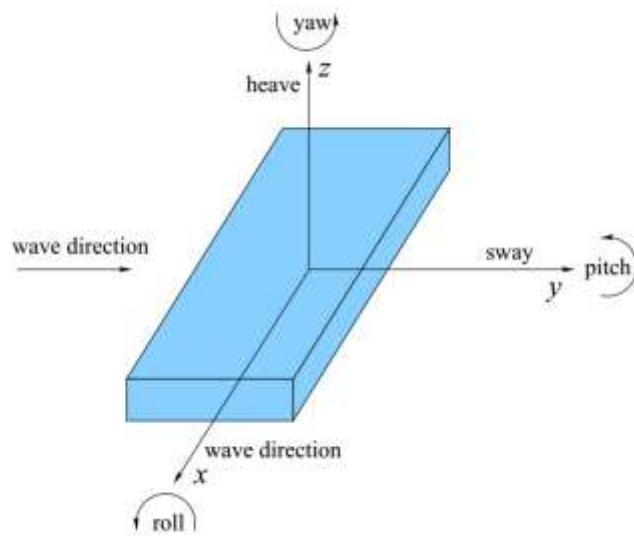


Figure 3. Coordinate system.

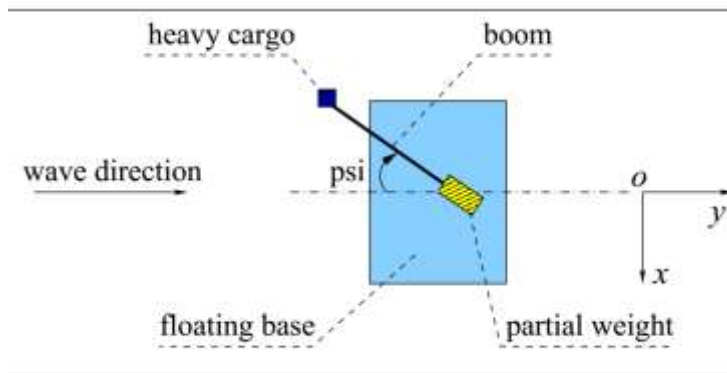
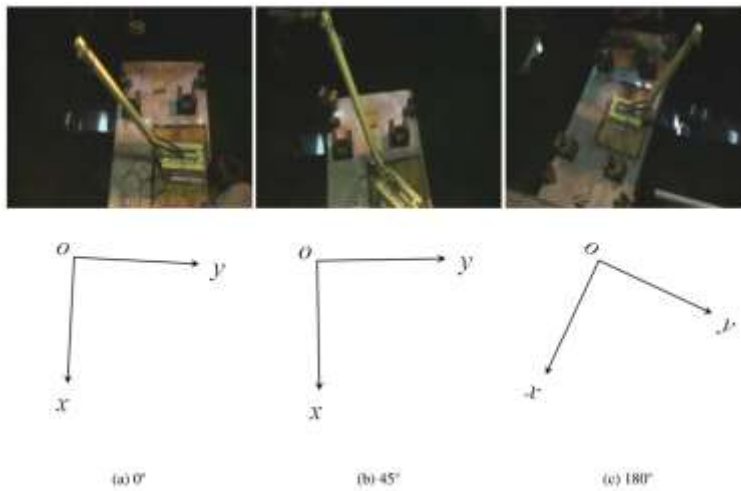
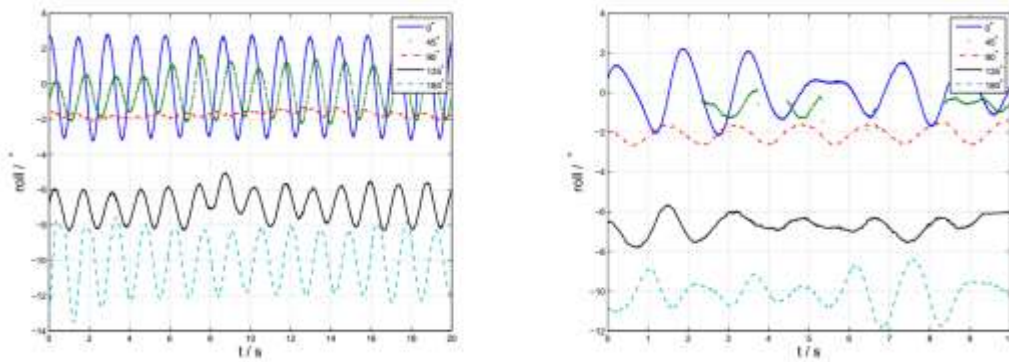


Figure 4. Top view sketch of the floating crane and the definition of boom angle.



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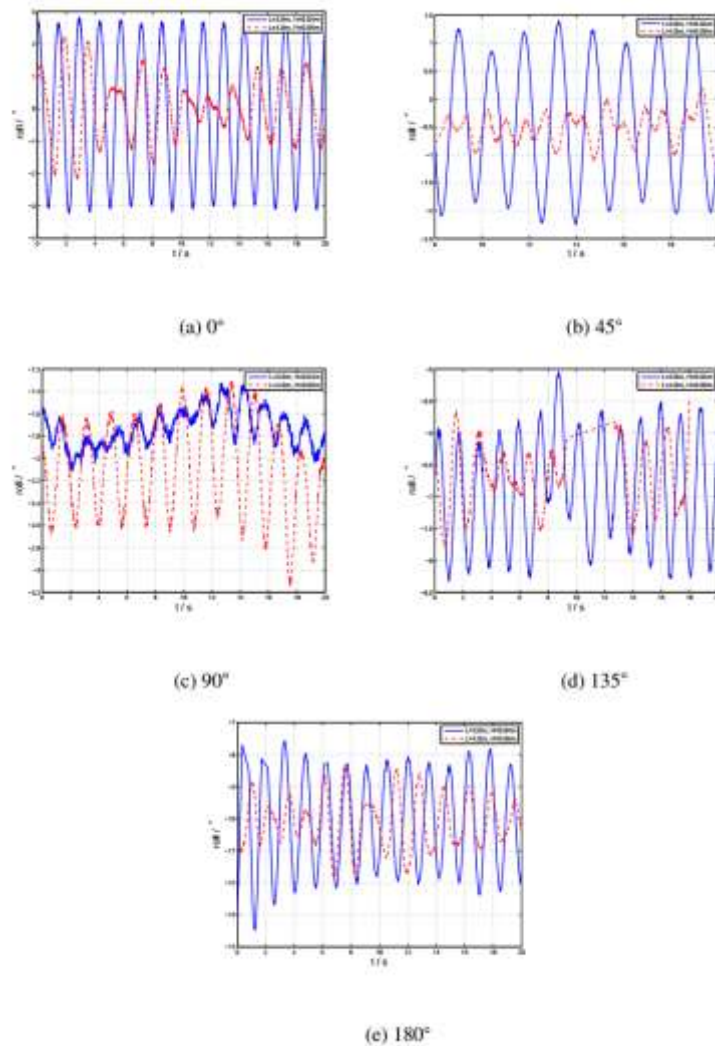
Figure 5. Snapshots of experimental layout at various boom angles.



(a) Case 1

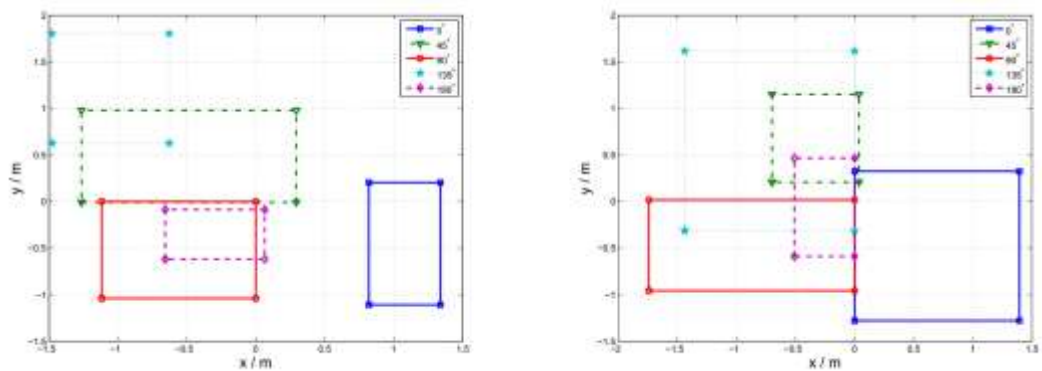
(b) Case 2

Figure 6. Time series of the roll motions of floating crane at various boom angles in two different incident wave conditions.



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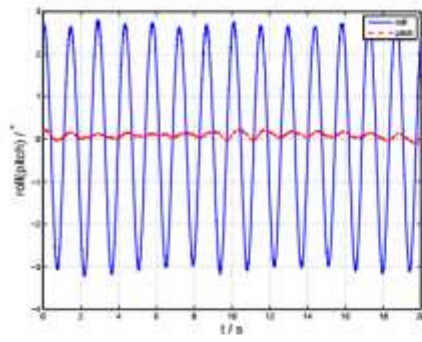
Figure 7. Time series of the roll motion at various boom angles.



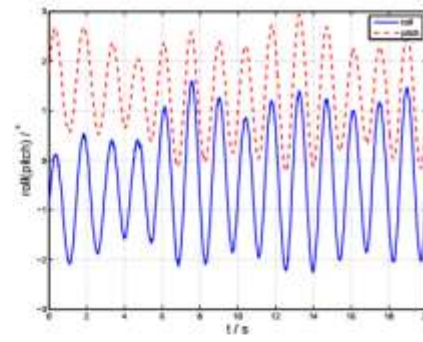
(a) Case 1

(b) Case 2

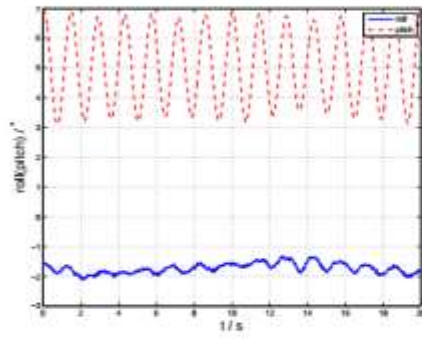
Figure 8. The projected area of the heavy cargo's motion



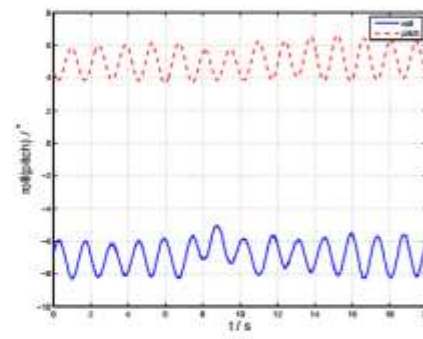
(a) 0°



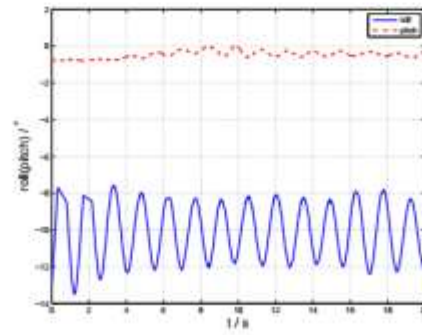
(b) 45°



(c) 90°

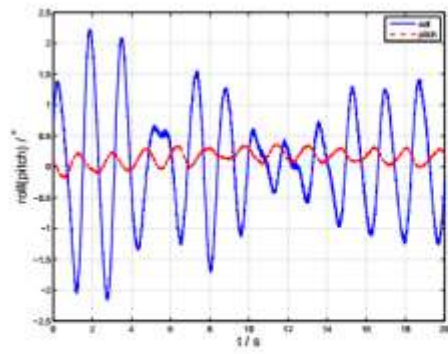


(d) 135°

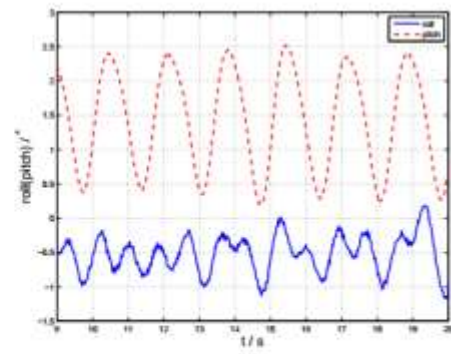


(e) 180°

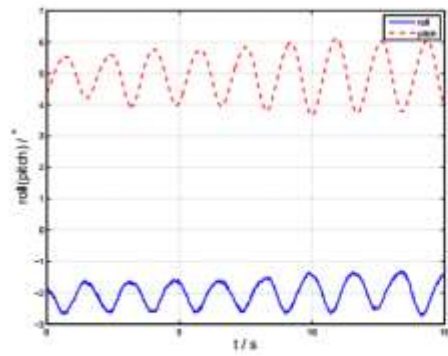
Figure 9. Roll and pitch motions in case 1 at different boom angles



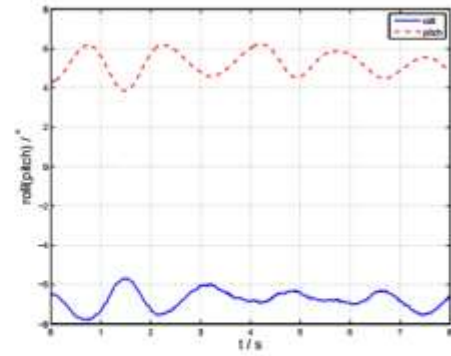
(a) 0°



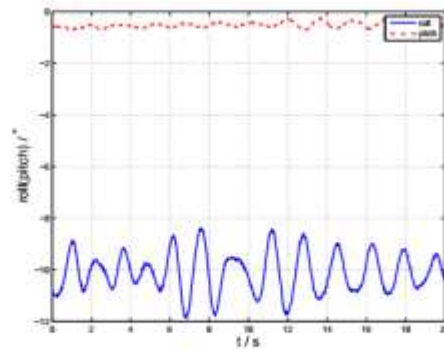
(b) 45°



(c) 90°



(d) 135°



(e) 180°

Figure 10. Roll and pitch motions in case 2 at different boom angles

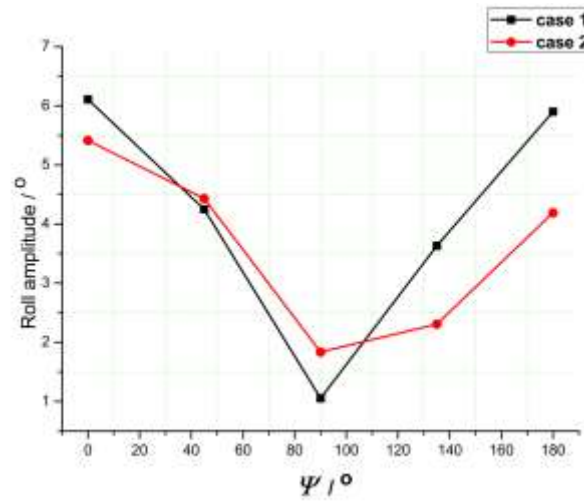


Figure 11. Roll amplitude at various boom angles

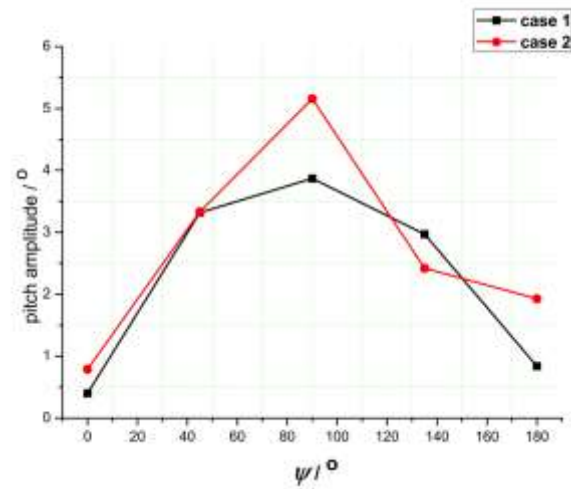


Figure 12. Pitch amplitude at various boom angles

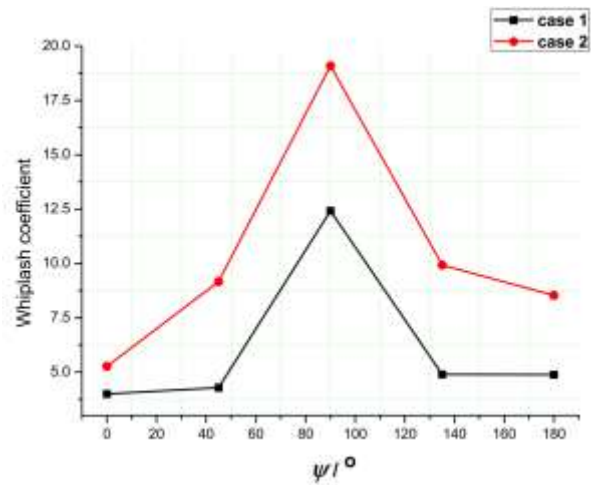


Figure 13. Roll whirlash coefficient at various boom angles

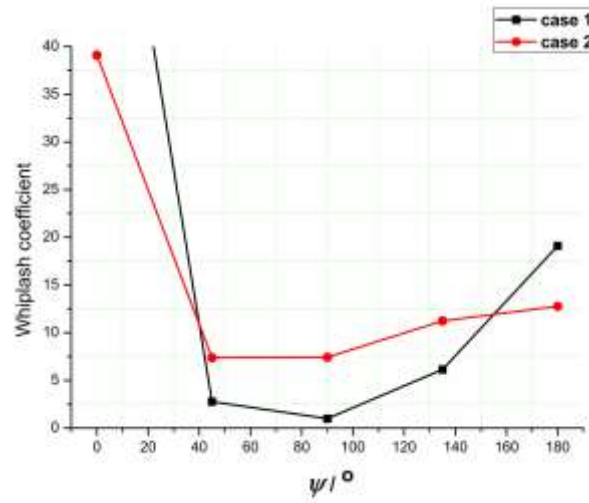


Figure 14. Pitch whiplash coefficient at various boom angles

Num.	Parts	Mass (kg)	Length, Width and Height (m)
①	Floating base	250	3, 1, 0.15
②	Partial weight	15	0.4, 0.1875, 0.05
③	Boom	5	0.8, 0.032, 0.032
④	Heavy cargo	10	0.136, 0.136, 0.136

Table 1. Physical parameters of the floating crane.